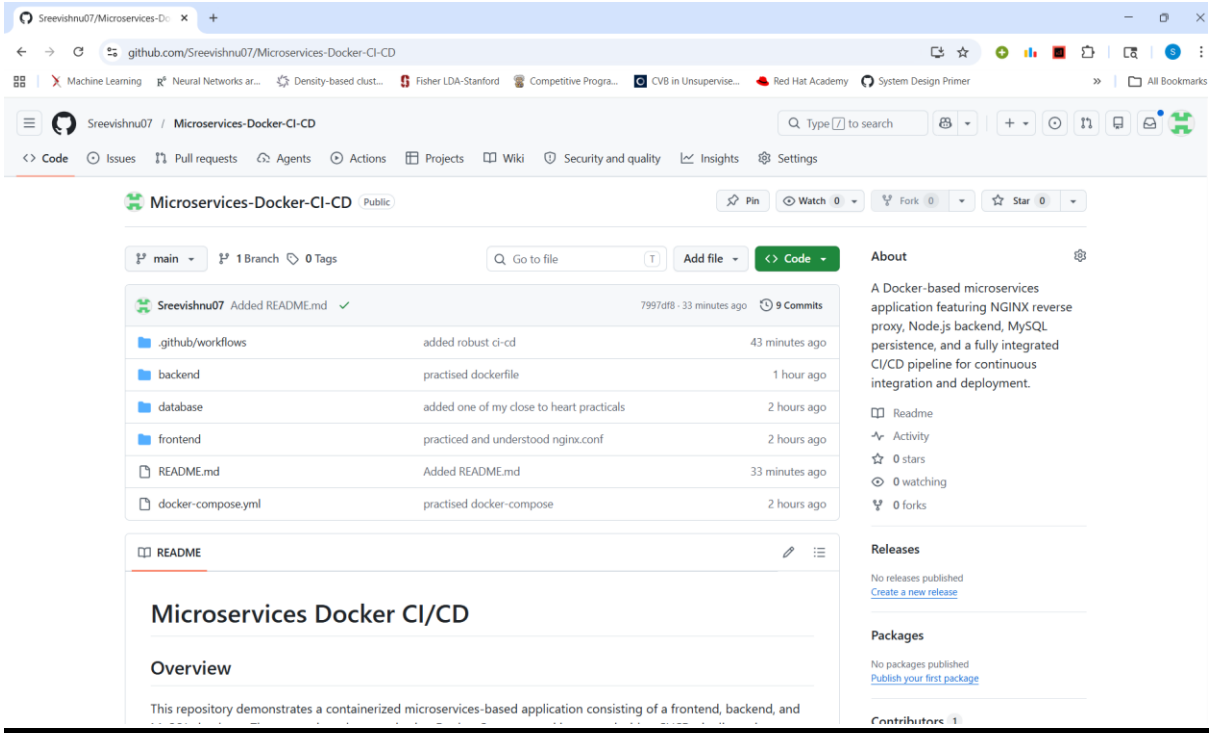
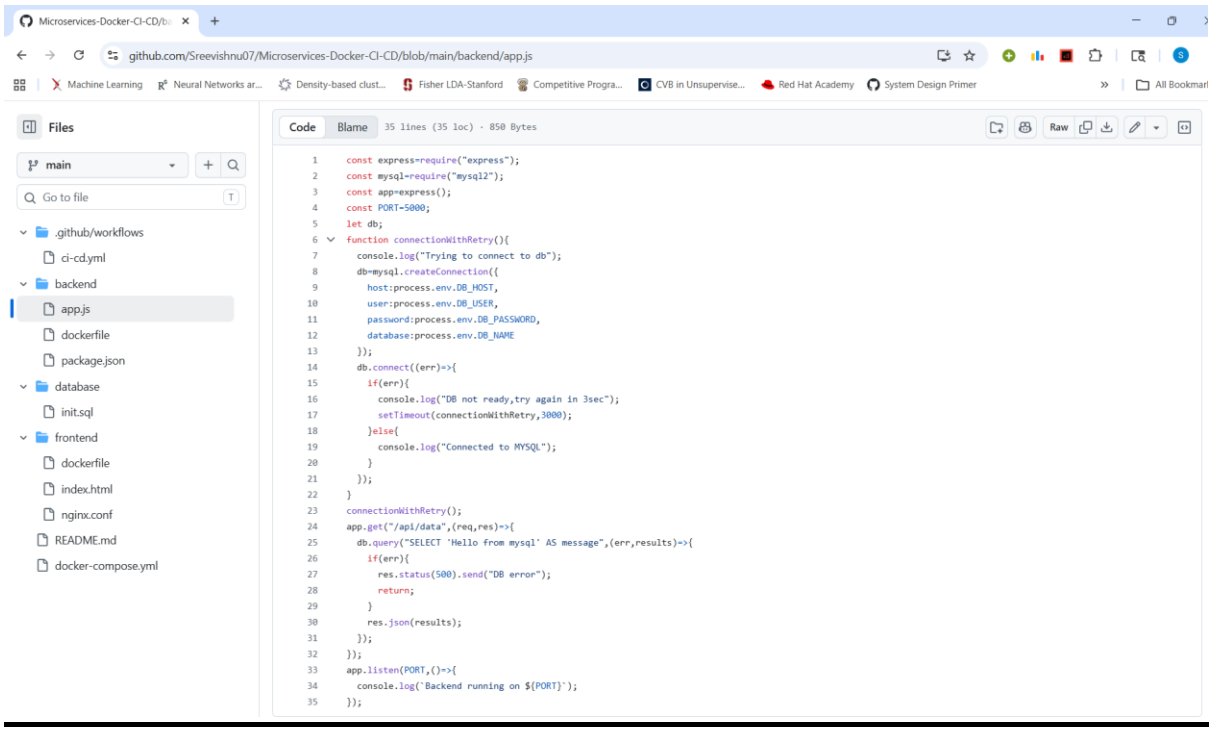
Design and Implementation of Docker Microservices with CI/CD Integration





github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/backend/dockerfile

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / backend / dockerfile

Sreevishnu07 practised dockerfile

Code | Blame | 7 lines (7 loc) · 103 Bytes

```
FROM node:18
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
EXPOSE 5000
CMD ["npm","start"]
```

---

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/backend/package.json

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmark

Sreevishnu07 / Microservices-Docker-CI-CD

Type / to search

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / backend / package.json

Sreevishnu07 added one of my close to heart practicals | 216ae3f · 3 hours ago | History

Code | Blame | 13 lines (13 loc) · 229 Bytes | Raw

```
{
"name":"backend-service",
"version":"1.0.0",
"description":"simple backend api for microservices demo",
"main":"app.js",
"scripts":{
  "start":"node app.js"
  },
"dependencies":{
  "express":"^4.18.2",
  "mysql2":"^3.6.0"
  }
}
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/dockerfile

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System I

Sreevishnu07 / Microservices-Docker-CI-CD

Type / to search

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / frontend / dockerfile

Sreevishnu07 added one of my close to heart practicals

Code | Blame | 5 lines (5 loc) · 142 Bytes

```
FROM nginx:latest
RUN rm /etc/nginx/conf.d/default.conf
COPY nginx.conf /etc/nginx/nginx.conf
COPY index.html /usr/share/nginx/html
EXPOSE 80
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/index.html

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

Sreevishnu07 / Microservices-Docker-CI-CD

Type / to search

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / frontend / index.html

Sreevishnu07 added one of my close to heart practicals

216ae3f · 3 hours ago | History

Code | Blame | 12 lines (12 loc) · 192 Bytes | Raw

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple microservices app</title>
</head>
<body>
  <h2>Click to get data</h2>
  <a href="/api/data">
     <button>Get data</button>
  </a>
</body>
</html>
```

Microservices-Docker-CI-CD/fr

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/frontend/nginx.conf

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy

Sreevishnu07 / Microservices-Docker-CI-CD

Type / to searc

Code | Issues | Pull requests | Agents | Actions | Projects | Wiki | Security and quality | Insights | Settings

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Microservices-Docker-CI-CD / frontend / nginx.conf

Sreevishnu07 practiced and understood nginx.conf

Code | Blame | 13 lines (13 loc) · 190 Bytes

```
events{}
http{
  server{
    listen 80;
    location /{
      root /usr/share/nginx/html;
      index index.html;
    }
    location /api/{
      proxy_pass http://backend:5000;
    }
  }
}
```

Microservices-Docker-CI-CD/d

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/docker-compose.yml

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

Files

main

Go to file

- .github/workflows
  - ci-cd.yml
- backend
  - app.js
  - dockerfile
  - package.json
- database
  - init.sql
- frontend
  - dockerfile
  - index.html
  - nginx.conf
- README.md
- docker-compose.yml

Code | Blame | 37 lines (35 loc) · 653 Bytes | Raw

```yaml
services:
  database:
    image: mysql:8
    container_name: mysqldb
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: mydb
    ports:
      - "3307:3306"
    volumes:
      - db_data:/var/lib/mysql

  backend:
    build: ./backend
    container_name: backend
    restart: always
    depends_on:
      - database
    environment:
      DB_HOST: database
      DB_USER: root
      DB_PASSWORD: root
      DB_NAME: mydb
    ports:
      - "5000:5000"

  frontend:
    build: ./frontend
    container_name: frontend
    restart: always
    depends_on:
      - backend
    ports:
      - "80:80"
```

Microservices-Docker-CI-CD/.g

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmarks

Files

main

Go to file

.github/workflows

ci-cd.yml

backend

database

frontend

README.md

docker-compose.yml

Sreevishnu07 Update ci-cd.yml

8686b4b · 3 minutes ago History

Code Blame 100 lines (75 loc) · 2.34 KB

Raw

```yaml
name: Full CI-CD Pipeline

on:
  push:
    branches:
      - main

jobs:
  test:
    runs-on: ubuntu-latest
    services:
      mysql:
        image: mysql:8
        ports:
          - "3306:3306"
        env:
          MYSQL_ROOT_PASSWORD: root
          MYSQL_DATABASE: my_db
        options: >-
          --health-cmd="mysqladmin ping -h localhost -proot"
          --health-interval=5s
          --health-timeout=5s
          --health-retries=5

    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Setup-node
        uses: actions/setup-node@v4
        with:
          node-version: 18
```

Microservices-Docker-CI-CD/.g

github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml

Machine Learning | Neural Networks ar... | Density-based clust... | Fisher LDA-Stanford | Competitive Progra... | CVB in Unsupervise... | Red Hat Academy | System Design Primer | All Bookmark

Files

main

Go to file

.github/workflows

ci-cd.yml

backend

database

frontend

README.md

docker-compose.yml

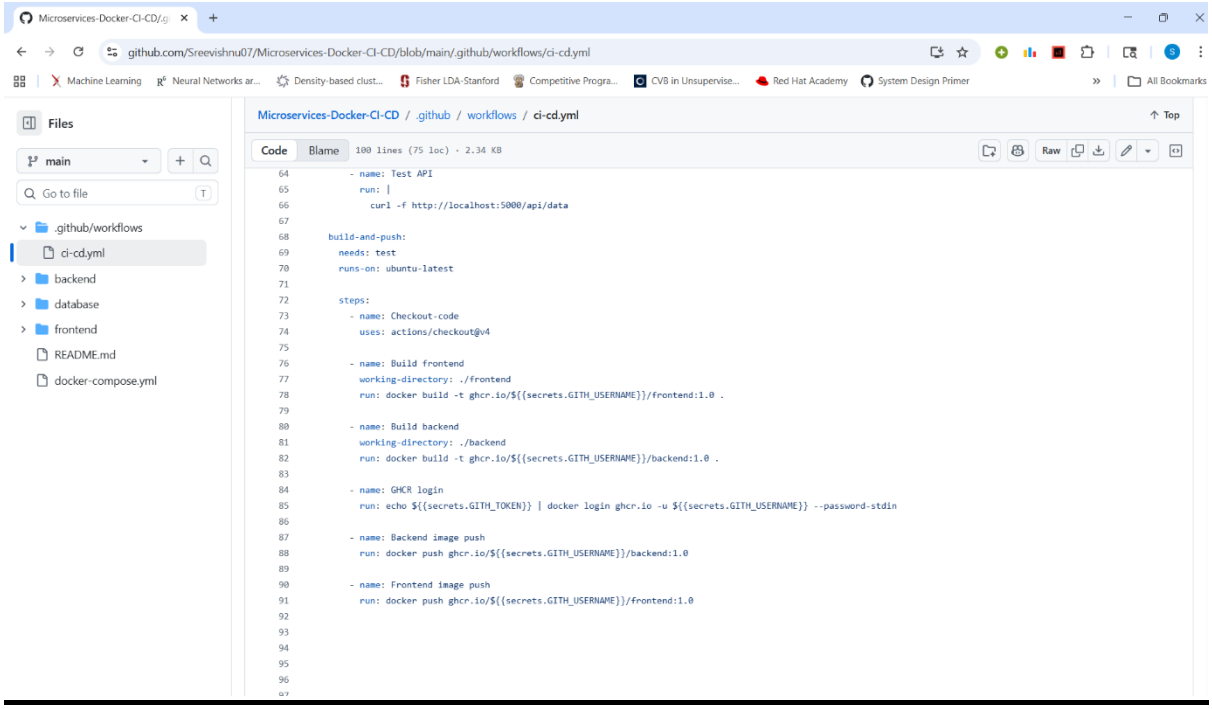
Microservices-Docker-CI-CD / .github / workflows / ci-cd.yml

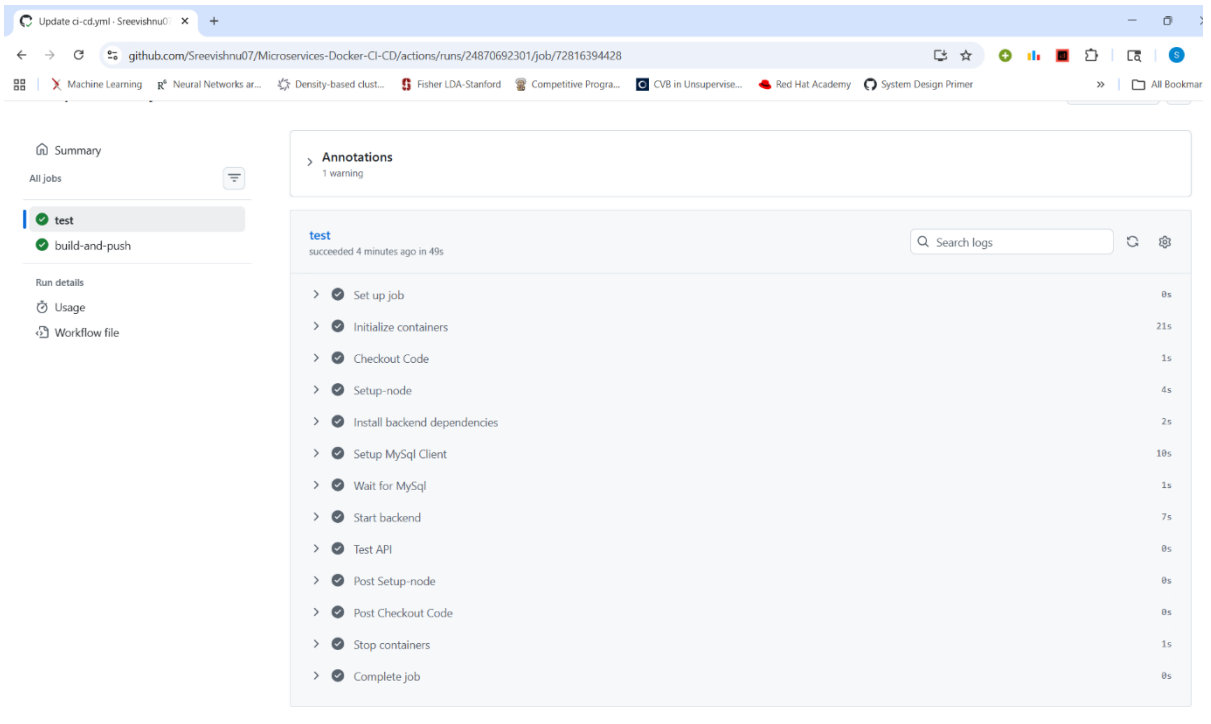
Top

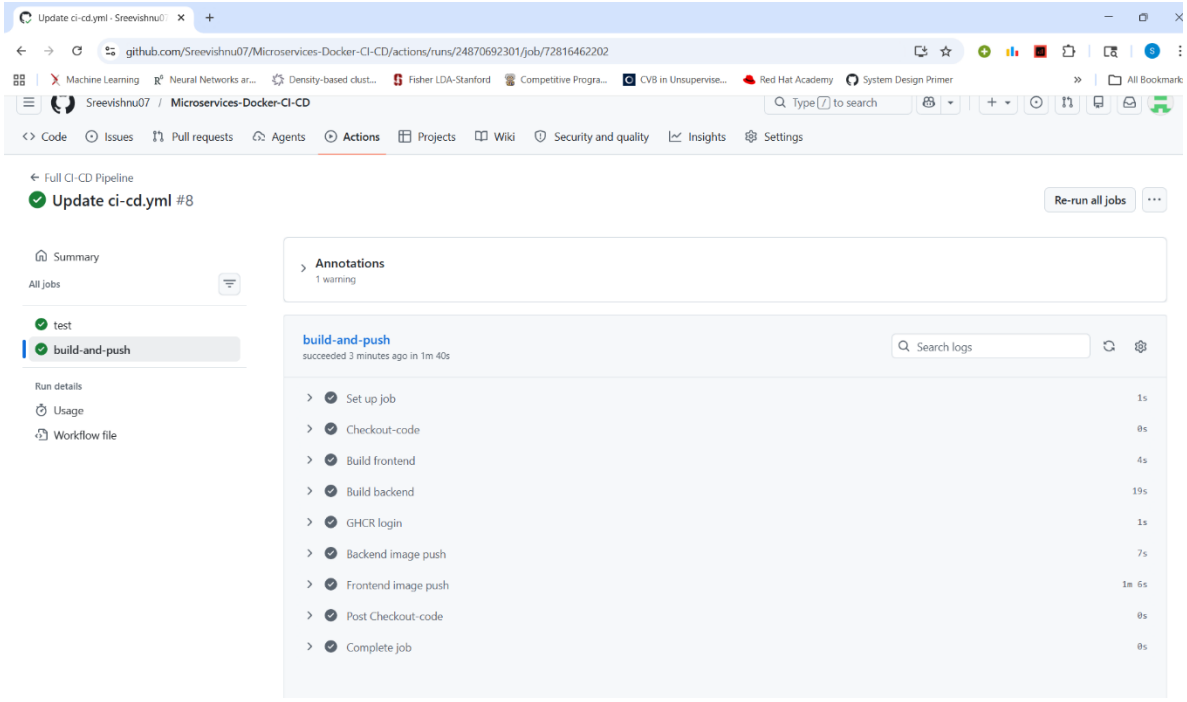
Code Blame 100 lines (75 loc) · 2.34 KB

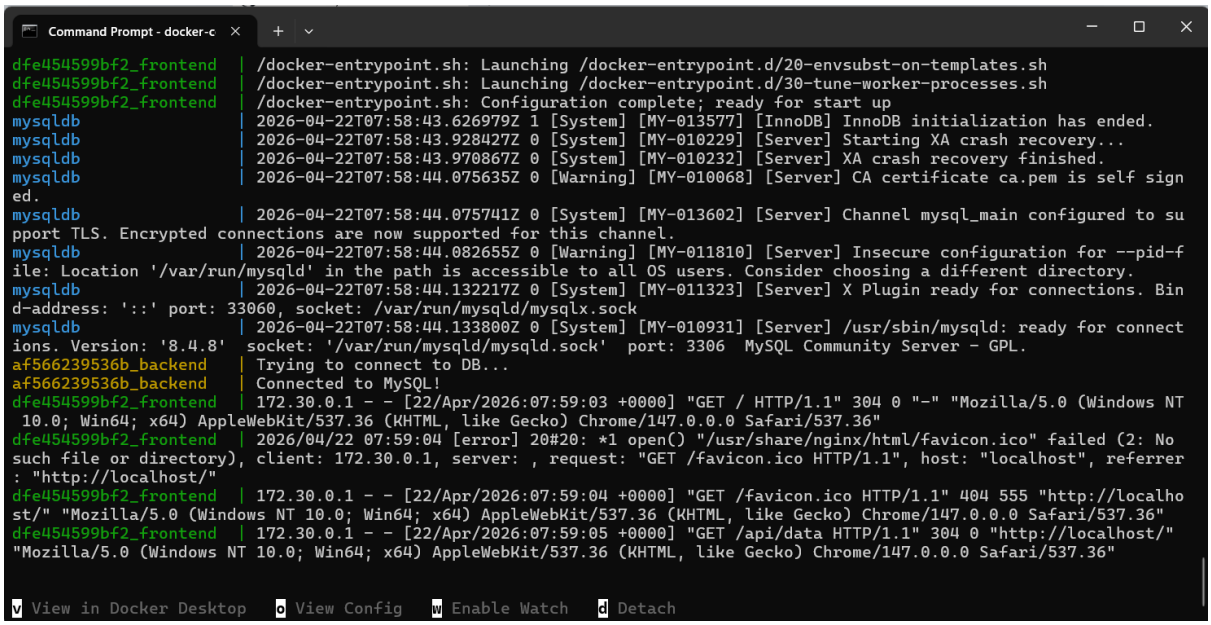
Raw

```yaml
      - name: Install backend dependencies
        working-directory: ./backend
        run: npm install

      - name: Setup MySql Client
        run: sudo apt-get update && sudo apt-get install mysql-client

      - name: Wait for MySql
        run: |
          for i in {1..10};do
            mysqladmin ping -h 127.0.0.1 -proot && break
            echo "waiting for mysql"
            sleep 3
          done

      - name: Start backend
        working-directory: ./backend
        env:
          DB_HOST: 127.0.0.1
          DB_USER: root
          DB_PASSWORD: root
          DB_NAME: my_db
        run: |
          node app.js &
          for i in {1..10}; do
            curl -f http://localhost:5000/api/data && break
            echo "Waiting for backend.."
            sleep 2
          done

      - name: Test API
        run: |
          curl -f http://localhost:5000/api/data

  build-and-push:
```

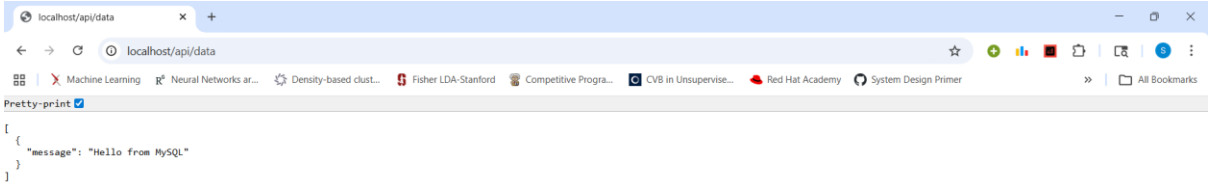

Microservices-Docker-CI-CD/.g
github.com/Sreevishnu07/Microservices-Docker-CI-CD/blob/main/.github/workflows/ci-cd.yml
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmarks
Files
main
Go to file
.github/workflows
ci-cd.yml
backend
database
frontend
README.md
docker-compose.yml
Microservices-Docker-CI-CD / .github / workflows / ci-cd.yml
Top
Code
Blame
100 lines (75 loc) · 2.34 KB
Raw
- name: Test API
run: |
curl -f http://localhost:5000/api/data
build-and-push:
needs: test
runs-on: ubuntu-latest
steps:
- name: Checkout-code
uses: actions/checkout@v4
- name: Build frontend
working-directory: ./frontend
run: docker build -t ghcr.io/${{secrets.GITH_USERNAME}}/frontend:1.0 .
- name: Build backend
working-directory: ./backend
run: docker build -t ghcr.io/${{secrets.GITH_USERNAME}}/backend:1.0 .
- name: GHCR login
run: echo ${{secrets.GITH_TOKEN}} | docker login ghcr.io -u ${{secrets.GITH_USERNAME}} --password-stdin
- name: Backend image push
run: docker push ghcr.io/${{secrets.GITH_USERNAME}}/backend:1.0
- name: Frontend image push
run: docker push ghcr.io/${{secrets.GITH_USERNAME}}/frontend:1.0


Update ci-cd.yml · Sreevishnu0
github.com/Sreevishnu07/Microservices-Docker-CI-CD/actions/runs/24870692301/job/72816394428
Machine Learning
Neural Networks ar...
Density-based clust...
Fisher LDA-Stanford
Competitive Progra...
CVB in Unsupervise...
Red Hat Academy
System Design Primer
All Bookmar
Summary
All jobs
test
build-and-push
Run details
Usage
Workflow file
Annotations
1 warning
test
succeeded 4 minutes ago in 49s
Search logs
Set up job 0s
Initialize containers 21s
Checkout Code 1s
Setup-node 4s
Install backend dependencies 2s
Setup MySql Client 10s
Wait for MySql 1s
Start backend 7s
Test API 0s
Post Setup-node 0s
Post Checkout Code 0s
Stop containers 1s
Complete job 0s


Update ci-cd.yml - Sreevishnu07
github.com/Sreevishnu07/Microservices-Docker-CI-CD/actions/runs/24870692301/job/72816462202
Sreevishnu07 / Microservices-Docker-CI-CD
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Full CI-CD Pipeline
Update ci-cd.yml #8
Re-run all jobs
Summary
All jobs
test
build-and-push
Run details
Usage
Workflow file
Annotations
1 warning
build-and-push
succeeded 3 minutes ago in 1m 40s
Search logs
Set up job 1s
Checkout-code 0s
Build frontend 4s
Build backend 19s
GHCR login 1s
Backend image push 7s
Frontend image push 1m 6s
Post Checkout-code 0s
Complete job 0s


Command Prompt - docker-c
dfe454599bf2_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
dfe454599bf2_frontend | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
dfe454599bf2_frontend | /docker-entrypoint.sh: Configuration complete; ready for start up
mysqldb | 2026-04-22T07:58:43.626979Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqldb | 2026-04-22T07:58:43.928427Z 0 [System] [MY-010229] [Server] Starting XA crash recovery...
mysqldb | 2026-04-22T07:58:43.970867Z 0 [System] [MY-010232] [Server] XA crash recovery finished.
mysqldb | 2026-04-22T07:58:44.075635Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self sign
ed.
mysqldb | 2026-04-22T07:58:44.075741Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to su
pport TLS. Encrypted connections are now supported for this channel.
mysqldb | 2026-04-22T07:58:44.082655Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-f
ile: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
mysqldb | 2026-04-22T07:58:44.132217Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bin
d-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysqldb | 2026-04-22T07:58:44.133800Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connect
ions. Version: '8.4.8' socket: '/var/run/mysqld/mysqld.sock' port: 3306 MySQL Community Server - GPL.
af566239536b_backend | Trying to connect to DB...
af566239536b_backend | Connected to MySQL!
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:03 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT
10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
dfe454599bf2_frontend | 2026/04/22 07:59:04 [error] 20#20: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No
such file or directory), client: 172.30.0.1, server: , request: "GET /favicon.ico HTTP/1.1", host: "localhost", referrer
: "http://localhost/"
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:04 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localho
st/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
dfe454599bf2_frontend | 172.30.0.1 - - [22/Apr/2026:07:59:05 +0000] "GET /api/data HTTP/1.1" 304 0 "http://localhost/"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36"
View in Docker Desktop
View Config
Enable Watch
Detach

Simple microservices app
localhost
Click to get data
Get data




```
[
  {
    "message": "Hello from MySQL"
  }
]
```